ASPHYS 7012 Stellar Physics HW 5

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1. The ionization energies of neutral helium and singly ionized helium are $\chi_I = 24.6\,eV$ and $\chi_{II} = 54.4\,eV$, respectively. The partition functions are $U_I = 1$, $U_{II} = 2$, and $U_{III} = 1$. Use electron pressure $P_e = 20\,Nm^{-2}$ for the following calculations.

(a) From Saha equation, find He_{II}/He_I and He_{III}/He_{II} at 5,000 K, 15,000 K, and 25,000 K. (3)

$$\frac{n_{r+1}}{n_r}P_e = \frac{u_{r+1}}{u_r}2\frac{(2\pi m_e)^{3/2}}{h^3}(kT)^{5/2}e^{-\frac{\chi_r}{kT}}$$

He_{II}/He_I at 5,000 K =

$$\frac{He_{II}}{He_I} = \frac{1}{P_e}\frac{U_{II}}{U_I}2\frac{(2\pi m_e)^{3/2}}{h^3}(5000k)^{5/2}e^{-\frac{\chi_I}{5000k}} \approx 1.887 \times 10^{-18}$$

He_{II}/He_I at 15,000 K =

$$\frac{He_{II}}{He_I} = \frac{1}{P_e}\frac{U_{II}}{U_I}2\frac{(2\pi m_e)^{3/2}}{h^3}(15000k)^{5/2}e^{-\frac{\chi_I}{15000k}} \approx 0.9976$$

He_{II}/He_I at 25,000 K =

$$\frac{He_{II}}{He_I} = \frac{1}{P_e}\frac{U_{II}}{U_I}2\frac{(2\pi m_e)^{3/2}}{h^3}(25000k)^{5/2}e^{-\frac{\chi_I}{25000k}} \approx 7239.052$$

He_{III}/He_{II} at 5,000 K =

$$\frac{He_{III}}{He_{II}} = \frac{1}{P_e}\frac{U_{III}}{U_{II}}2\frac{(2\pi m_e)^{3/2}}{h^3}(5000k)^{5/2}e^{-\frac{\chi_{II}}{5000k}} \approx 4.331 \times 10^{-49}$$

He_{III}/He_{II} at 15,000 K =

$$\frac{He_{III}}{He_{II}} = \frac{1}{P_e}\frac{U_{III}}{U_{II}}2\frac{(2\pi m_e)^{3/2}}{h^3}(15000k)^{5/2}e^{-\frac{\chi_{II}}{15000k}} \approx 2.424 \times 10^{-11}$$

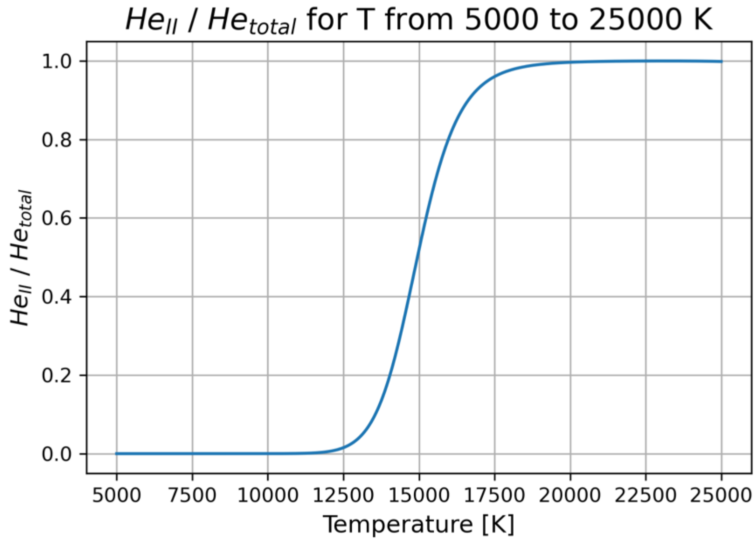
He_{III}/He_{II} at 25,000 K =

$$\frac{He_{III}}{He_{II}} = \frac{1}{P_e}\frac{U_{III}}{U_{II}}2\frac{(2\pi m_e)^{3/2}}{h^3}(25000k)^{5/2}e^{-\frac{\chi_{II}}{25000k}} \approx 0.00178$$

(b) Express $He_{II}/(He_I + He_{II} + He_{III})$ in terms of He_{II}/He_I and He_{III}/He_{II}. (2)

$$\frac{He_{II}}{He_I + He_{II} + He_{III}} = \frac{1}{\frac{He_I}{He_{II}} + 1 + \frac{He_{III}}{He_{II}}} = \frac{1}{\frac{1}{\frac{He_{II}}{He_I}} + 1 + \frac{He_{III}}{He_{II}}}$$

(c) $He_{total} = He_I + He_{II} + He_{III}$. Plot He_{II}/He_{total} for a range of temperatures from 5,000 K to 25,000 K. You can either write a program to plot it or draw by hand by calculating He_{II}/He_{total} at various temperatures and then connecting the dots. (4)



(d) What is the temperature that $He_{II}/He_{total} \simeq 0.5$? I ask for precision down to hundreds (put 0 for tens and ones). (3)

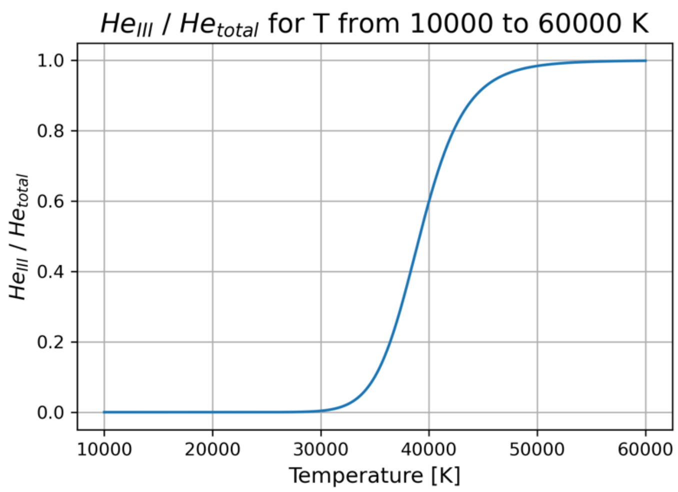
The temperature that $He_{II}/He_{total} \simeq 0.5$ is about 14900 K.

2. Following Q1, use election pressure $P_e = 1000\,\mathrm{Nm}^{-2}$.

(a) Express He_{III}/He_{total} in terms of He_{II}/He_{I} and He_{III}/He_{II} . (2)

$$\frac{He_{III}}{He_I + He_{II} + He_{III}} = \frac{\frac{He_{III}}{He_{II}}}{\frac{He_I + He_{II} + He_{III}}{He_{II}}} = \frac{\frac{He_{III}}{He_{II}}}{\frac{1}{\frac{He_{II}}{He_I}} + 1 + \frac{He_{III}}{He_{II}}}$$

(b) Plot He_{III}/He_{total} for a range of temperatures from 10,000 K to 60,000 K. (4)



(c) What is the temperature that $He_{III}/He_{total} \simeq 0.5$? (with precision down to thousands.) (3)

The temperature that $He_{III}/He_{total} \simeq 0.5$ is about 39100 K.

3. Figure 1 shows the spectra of two stars of the same spectral type but different luminosity classes.
   (a) Which one has a higher luminosity? (2)
   The upper spectra of star.
   (b) Why?
   The lower spectra of star has stronger absorption line of Hγ, Hδ, Hε ,Hg. Under certain temperature conditions (lower temperature), there will be strong metal absorption lines.